MIL-P-63357A (AR)
31 March 1980
SUPERSEDING
MIL-P-63357 (AR)
3 August 1979

MILITARY SPECIFICATION

PROJECTILE, TRAINING, 155MM, M823

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not covered by the drawings and provides quality assurance provisions for the fabrication of parts, assembly and packaging of a training projectile, M823, to be used to simulate the handling characteristics of the M712, a 155mm Cannon Launched Guided Projectile (Copperhead).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-C-490 -Cleaning Methods and Pretreatment of Surfaces For Organic Coatings

MILITARY

MIL-A-8625 -Anodic Coatings, For Aluminum and Aluminum Alloys

MIL-A-48078 -Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-810 Environmental Test Methods

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC: 1320

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DRAWINGS

9329721	-Projectile, 155mm, Training, M823 Assembly (Inert)
9305335	-Packaging Assembly for Projectile, 155mm, Guided, Cannon, Launched M712 Assembly and Training M823 with Packing and Marking instructions
9305425	-Container, Metal, for Projectile 155MM, Guided, Cannon, Launched, M712 and Trainer M823
9305435	-6-Round Copperhead Pallet Unit.
9305465	-Extractor Tool Assembly

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials - Materials shall be in accordance with the applicable drawings, specifications and standards.

3.2 Components and Assemblies - The components and assemblies shall comply with all requirements specified on drawing 9331967 and associated drawings and with all requirements specified in applicable specifications and standards.

3.2.1 Obturator - The obturator (dwg. 9332456/9308030), when assembled as shown on the trainer assembly (dwg. 9331967) shall be capable of being rotated by hand.

3.3. Ram and Extract - The assembled training projectile (9329721) shall be capable of being hand or power rammed and seated in either a M185 or M199 howitzer tube. The projectile shall also be capable of being removed from the howitzer by using extractor (9305465).

3.4 Transportation Vibration - The containerized training projectile shall withstand a sinusoidally cycled vibration ranging from one inch double amplitude at 5 Hertz frequency to 0.0017 inch double amplitude at 200 Herz frequency and shall remain intact as evidenced by loose or damaged components.

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3.5 First Article Inspection - This specification contains technical provisions for first article inspection. The first article sample shall meet all requirements of this specification. Requirements for the submission of first article sample by the contractor shall be as specified in the contract.

3.6 Workmanship - The parts and assemblies shall be free from chips, burrs, gouges, cracks, dirt, grease and other foreign matter that may impair proper functioning. Cleaning methods used shall not be injurious to any of the parts, nor shall the parts be contaminated.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections - The following types of inspection shall be conducted on this item.

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission - The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Projectile, 155mm, Training, M823 Assembly (Inert)	9329721	4
Coating Test Panels	See Table I	12

4.3.2 Inspections to be Performed - See MIL-A-48078 and Table I specified herein. The special tests per paragraphs 4.5.1 and 4.5.2 shall be performed by the Government at a Government FAcility to be designated by the contracting officer.

4.3.3 Rejection - See Mil-A-48078.

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TABLE 1 - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Projectile 155MM, Training, M823 Assembly (Inert)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Projectile 155MM, Training, M823 Assembly (Inert) dwg 9329721	2		3.2	Gage/Visual (4.4.2.1 - 4.4.2.10)
	Projectile 155MM, Training, M823 Assembly (Inert) dwg 9329721	1		3.3	Test (4.5.1)
	Projectile 155MM, Training, M823 Assembly (Inert) dwg 9329721	1		3.4	Test (4.5.2)

NOTES:

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TABLE 1 - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	Projectile, 155MM, Training, M823 Assembly (Inert)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Body, Forward				
	dwg 9331972				
	weight of coating	3*		3.2	4.5.3
	Body, Assembly				
	dwg 9331160				
	weight of coating	3*		3.2	4.5.4
	corrosion resistance of coating	3*		3.2	4.5.4
	light fastness resistance of coating	3*		3.2	4.5.4

NOTES:

* Test panels.

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4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation - Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination - See MIL-A-48078.

a. Sampling Plans - Unless otherwise specified in the Clasification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63357A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Fin		9331159
			NEXT HIGHER ASSEMBLY 9331160

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR	None defined				
MINOR:					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63357A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Switch, Rotary		9332454
			NEXT HIGHER ASSEMBLY 9331768

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None				
MAJOR	None				
MINOR					
201	Rotational Torque (min.)		0.65%	3.2	Gage
202	Rotational Torque (max.)		0.65%	3.2	Gage
203	Switch Positions positive		0.65%	3.2	Visual/Manual
204	Switch Mechanical Life	2/Lot		3.2	Test
205	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63357A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9331768
4.4.2.3	Switch Assembly				NEXT HIGHER ASSEMBLY 9331967
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR	None defined				
MINOR					
201	Components missing or loose		0.65%	3.2	Visual
202	Metal components not properly finished		0.65%	3.2	Visual
203	Numbers on screw slot missing from Rotary Switch Knob		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63357A (AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.4	Base	1	1	9331769
				NEXT HIGHER ASSEMBLY 9331967

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None				
MAJOR					
101	Outside diameter of shoulder		0.40%	3.2	Gage
102	Pitch diameter of thread		0.40%	3.2	Gage
103	Major diameter of thread, min.		0.40%	3.2	Gage
104	Diameter of groove in Boattail		0.40%	3.2	Gage
105	Width of groove in Boattail		0.40%	3.2	Gage
MINOR					
201	Outside diameter		0.65%	3.2	Gage
202	Diameter of Boattail		0.65%	3.2	Gage
203	Length of base		0.65%	3.2	Gage
204	Protective finish improper		0.65%	3.2	Gage
205	Evidence of poor workmanship		0.65%	3.6	Visual
206	Hardness (2 places)		0.65%	3.2	ASTM E-8

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63357A (AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.5	Body, Forward	1	1	9331972
				NEXT HIGHER ASSEMBLY 9331967

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None				
MAJOR					
101	Pitch diameter of Whitworth thread		0.40%	3.2	Gage
102	Major diameter of Whitworth thread, min.		0.40%	3.2	Gage
103	Length of Whitworth thread		0.40%	3.2	Gage
104	Pitch diameter of large thread		0.40%	3.2	Gage
105	Major diameter of large thread, min.		0.40%	3.2	Gage
106	Length of large thread		0.40%	3.2	Gage
MINOR					
201	Outside diameter		0.65%	3.2	Gage
202	Length		0.65%	3.2	Gage
203	Protective finish improper		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63357A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.6	Body, Main		9331973
			NEXT HIGHER ASSEMBLY 9331160

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None				
MAJOR					
101	Pitch diameter of thread (2 places)		0.40%	3.2	Gage
102	Minor diameter of thread, max. (2 places)		0.40%	3.2	Gage
103	Length of thread (2 places)		0.40%	3.2	Gage
104	Position of thread to bourrelet		0.40%	3.2	Gage
105	Diameter of bourrelet		0.40%	3.2	Gage
106	Length of body		0.40%	3.2	Gage
MINOR					
201	Width of switch slot		0.65%	3.2	Gage
202	Length of switch slot		0.65%	3.2	Gage
203	Depth of switch slot		0.65%	3.2	Gage
204	Fin Grooves Missing or obviously improper		0.65%	3.2	Visual
205	Four shallow holes missing or improper		0.65%	3.2	Visual
206	Evidence of poor workmanship		0.65%	3.6	Visual
207	Runout of fwd Bourrellet with aft Bourrellet		0.65%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P- 63357A (AR)

PARAGRAPH	TITLE		SHEET	OF	DRAWING NUMBER
4.4.2.7	Window		1	1	9332455*
					NEXT HIGHER ASSEMBLY: 9331967

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Pitch diameter of thread		0.40%	3.2	Gage
102	Minor diameter of thread, max.		0.40%	3.2	Gage
103	Length of thread		0.40%	3.2	Gage
MINOR					
201	Length of window		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

* 9306095 optional design

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63357A (AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.8	Obturator	1	1	9332456 *
				NEXT HIGHER ASSEMBLY: 9331967

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Outside diameter		0.40%	3.2	Gage
102	Inside diameter		0.40%	3.2	Gage
MINOR					
201	Length		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

* 9308030 optional design

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63357A (AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.9	Trainer Assembly	1	1	9331967
				NEXT HIGHER ASSEMBLY 9329721

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Length		0.40%	3.2	Gage
102	Bourrelet diameter max, (3 places)		100%	3.2	Gage
103	Torque of main body to base		0.40%	3.2	Gage
104	Center of gravity improper		0.40%	3.2	Test
105	Weight		0.40%	3.2	Test
106	Feeler gage check		0.40%	3.2	Gage
107	Free rotation of obturator		0.40%	3.2	Manual
108	One cycle of rotary switch		0.40%	3.2	Manual
109	Any components missing, loose, or improperly assembled		100%	3.2	Visual/Manual
MINOR					
201	Evidence of poor workmanship		0.65%	3.6	Visual
202	Protective finish improper on any component		0.65%	3.2	Visual

NOTES:

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PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.10	Projectile, 155mm, Training, M823 Assembly	1	1	9329721
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR	None defined				
MINOR					
201	Marking improper		0.65%	3.2	Visual
202	Color improper		0.65%	3.2	Visual
203	Protective finish missing over marking		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63357A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Packaging Assembly	1 OF 1	9305335
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR	None defined				
MINOR					
201	Marking improper		0.65%	5.1	Visual
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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4.4.3 Testing - Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.4 Inspection Equipment - The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and section 6 herein.

4.5 Test Methods and Procedures

4.5.1 Ram and Extract Test - Projectile shall be rammed, seated and extracted one time using a M199 or M185 cannon tube, and the extractor (9305465)

4.5.2 Transportation Vibration Test - The containerized projectile shall be vibrated in accordance with MIL-STD-810, figures 514.2-7 curve AX and 514.2-10 except search and dwell tests shall be omitted. Tests shall be conducted in three axes.

4.5.3 Body, Forward Coating Test - Test panels prepared in accordance with TT-C-490 shall be tested for coating weight in accordance with TT-C-490, paragraph 4.2.6.

4.5.4 Body, Assembly Coating Test - Test panels prepared in accordance with MIL-A-8625 shall be tested for coating weight, coating corrosion resistance and light fastness in accordance with MIL-A-8625.

5. PACKAGING

5.1 Packing, Packaging and Marking - Packing and packaging of the projectile and marking of the loaded container shall be in accordance with drawing 9305335, part no. 9305335-002.

5.1.1 Container - Projectile container shall be in accordance with drawing 9305425.

5.2 Palletization - Packaged projectiles shall be palletized for shipment in accordance with drawing 9305435.

6. NOTES

6.1 Intended Use - The training projectile covered by this specification is intended to be used to simulate the handling characteristics of the M712 Cannon Launched Guided Projectile (Copperhead).

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6.2 Ordering Data - See MIL-A-48078.

6.3 Submission of Inspection Equipment for Design Approval - See MIL-A-48078. Submit designs as required to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, N.J. 07801.

Custodian
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Preparing Activity:
Army-AR

Project Number: 1320-A520

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